Curriculum Vitae



Maged Badawi
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PROFESSIONAL & TECHNICAL EXPERIENCE (~3 years 6 months):

Biomedical Engineer (~2 years 9 months):

Siemens Healthineers - Erlangen, Germany — May 2023 – Feb 2026

- Contributed to the development of **AR/VR**-based digital education content for **medical equipment**, enhancing interactive learning experiences.
- Contributed to product and hardware lifecycle **management**, including strategic planning, backlog tracking, and close coordination with **multidisciplinary** teams.
- Authored detailed technical documentation, designed **Power BI** dashboards for product insights, and led live demonstrations to showcase key features and training workflows to stakeholders.

AI, Software & Biomedical Engineering Projects:

MedTwin (Medical Equipment Digital Twin & SaaS Platform)

Next.js, TypeScript, React, Supabase, PostgreSQL, Vercel, GitHub, REST APIs. — Oct.-Nov 2026

- Developed a cloud-based SaaS platform for medical equipment management, maintenance, work orders, and multi-facility operations.
- Implemented secure multi-tenant architecture, role-based access control, authentication, and PostgreSQL RLS.
- Built equipment digital profiles, operational dashboards, and a centralized administration system.

AI-Powered System Design & Agentic Architecture

TypeScript, LangGraph.js, AI Agents/LLMs, REST APIs, Mermaid, Zod, Zustand, Git, Database Integration. — Mar.-June 2026

- Developed an AI-assisted workflow that transforms user requirements into structured system specifications and architecture diagrams.
- Integrated AI agents, backend services, and interactive frontend components for requirements analysis and automated diagram generation.

AI-Powered Knowledge Workspace Platform

Next.js, TypeScript, React, FastAPI, Python, spaCy, GitHub — Jan-Mar. 2026

- Developed a full-stack knowledge management platform with interactive node-based editing, structured information management, and multilingual AI processing.
- Built AI agents for entity extraction, relationship analysis, and knowledge structuring, integrating backend APIs with the frontend.

Fully Local RAG Assistant

FastAPI, Uvicorn, Pydantic, LangChain, FAISS, Ollama (phi3:mini), Docker, Git — Jan 2026

- Developed a local PDF question-answering system with automated document ingestion, vector similarity search, and offline LLM inference.
- Containerized the application using Docker Compose and implemented GitHub Actions CI/CD workflows.

Chest X-Ray Synthetic Image Generator using VAE

Python, PyTorch, CNN, Docker, HDF5 — Oct 2025

- Developed a Variational Autoencoder (VAE) to generate synthetic chest X-ray images for medical imaging research.
- Implemented a containerized training pipeline using Docker and HDF5 datasets.

Transformer-Based Text Classification

PyTorch, NLP, Transformers, Python — Sept. 2025

- Developed a transformer-based model for spam comment detection, including tokenization, attention mechanisms, and model evaluation.

Artificial Neural Network-Based Volume Conductor Solver for Electrical Spinal Cord Stimulation Applications

Master Thesis - Sim4Life, Python, DeepXDE, GNN, 3DCNN, PDE Modeling — Aug. 2025

- Built an automatic 3D voxel-based data generation pipeline using Sim4Life, creating a custom spinal cord dataset with tissue conductivity mapping, anatomical segmentation, and stimulation electrode placements using Dirichlet/Neumann boundary conditions.
- Implemented and trained neural network models to predict electric potential propagation in biological tissues.
- Performed benchmarking against Finite Element Method (FEM) simulations from Sim4Life, and reduced computational cost & inference time.

Deep Learning Classification of X-Ray Images (Normal, Pneumonia, Tuberculosis)

Python, CNN, PyTorch, ImageNet Transfer Learning — 2024

- Trained CNN models to classify X-ray medical images across multi-label disease categories.
- Applied data preprocessing, augmentation, and model fine-tuning to enhance classification accuracy.

Field Service Engineer (Full-Time, ~6 months**):**
AXA Medical - Cairo, Egypt Sept. 2021 – Mar. 2022

- Performed precise installation and configuration of medical devices in compliance with manufacturer protocols and clinical standards.
- Diagnosed and resolved hardware and software issues through systematic troubleshooting, minimizing downtime and ensuring operational efficiency.
- Conducted regular preventive and corrective maintenance on ICU and OR equipment, ensuring reliability and adherence to hospital safety regulations.
- Provided comprehensive training to medical staff on the safe and effective use of medical technologies, improving clinical workflow and device utilization.

Service & Clinical Engineering Intern (~3.5 months)
Cairo & Mansoura, Egypt | Jeddah, Saudi Arabia 2018 – 2021

EDUCATION:

Master's degree in Medical Engineering (Medical Image and Data Processing):
University of Erlangen-Nuremberg (FAU) – Erlangen, Germany 2022 - 2025

- **Dissertation**: **Artificial Neural Network**-based **Volume Conductor Solver** for Electrical Spinal Cord Stimulation Applications.

Bachelor's degree in Systems and Biomedical Engineering (5-year degree):
The higher Institute of Engineering in Al-Shorouk City (SH.A.) – Cairo, Egypt 2016 - 2021

- **Dissertation:** A medical mobile application for addressing the educational challenges by stimulating human memories using augmented reality technology developed by Unity engine in an interactive way.

Ideal Student Title:
The higher Institute of Engineering in Al-Shorouk City (SH.A.) - Cairo Egypt 2021

CERTIFICATIONS:

Medical Device Regulation.	Sept. 2024
Applications of Nanotechnology in Cardiovascular Diseases.	Jan. 2023
Python and Raspberry Pi Workshop.	Jun. 2019 - Jul. 2019
Analog and Digital Workshop.	Jun. 2018 - Jul. 2018
MATLAB.	Jul. 2018
Arduino Workshop.	Oct. 2017 - Feb. 2018

TECHNICAL SKILLS & SCIENTIFIC KNOWLEDGE:

Programming & Software Development: Python (intermediate–advanced), MATLAB & Simulink (intermediate–advanced), C, C++, C# (intermediate), HTML, CSS.

AI & Automation: Strong background in Artificial Intelligence, Machine Learning, and Deep Learning (AI/ML/DL); AI Agents, Large Language Models (LLMs) and RAG. PyTorch, TensorFlow, and SciPy. n8n.

Data Analysis & Visualization: Power BI, Power Automate, SQL, Numpy, Pandas, SciPy, Matplotlib, Seaborn, Pyvista

DevOps, Cloud & CI/CD: Git, GitHub, GitHub Actions (CI/CD), Docker (containerization), Microsoft Azure Fundamentals (AZ-900).

Embedded Systems & Electronics: Embedded systems programming using C with PIC microcontrollers and Python with Raspberry Pi; electronic and electrical engineering fundamentals; Analog and digital circuits; Proteus for electrical and electronic circuit design and simulation.

PUBLICATIONS & RESEARCH PROJECTS:

Rehabilitative Game-based System for Enhancing Physical and Cognitive Abilities of Neurological Disorders.	Mar. 2025
Review of **Zero-Shot** and **Few-Shot AI Algorithms** in The Medical Domain.	Jun. 2024
Flat-panel **CT** reconstruction.	Jul. 2023
Deep learning fully connected layers **model**.	Oct. 2022

LANGUAGES:

Arabic: Native.
English: Professional Proficiency.
German: Intermediate Proficiency.
French: Elementary Proficiency.